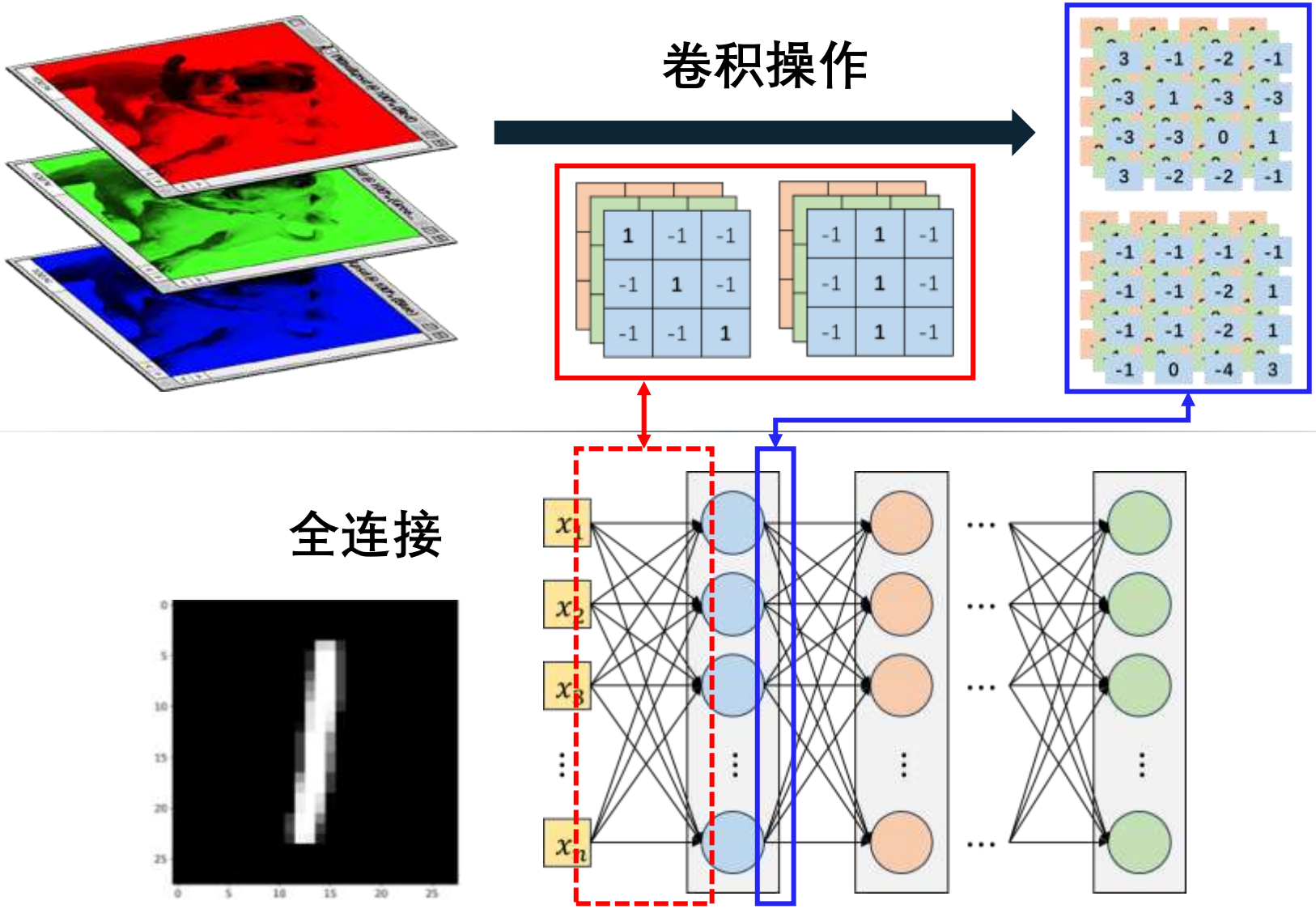
第6.2节 AlexNet

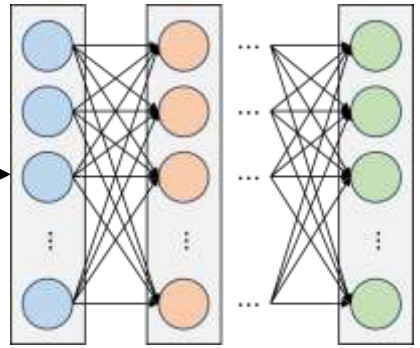
LeNet & AlexNet & ResNet

前情回顾：卷积神经网络 vs 全连接神经网络



前情回顾：卷积神经网络解决之问题

卷积层 → 池化层 → 卷积层 → 池化层 →



Classification

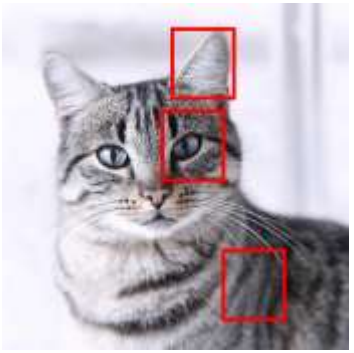
CAT

Classification + Localization

CAT

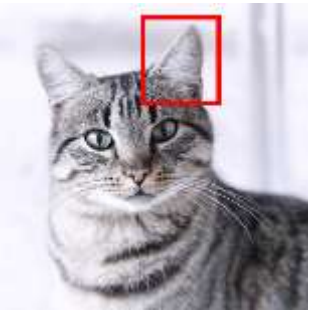
特点 1

局部特征远小于整幅图像



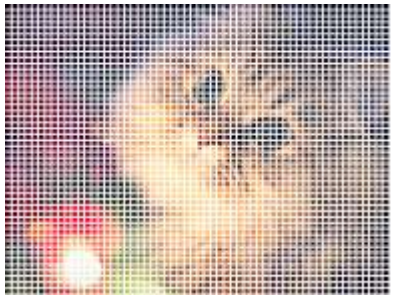
特点 2

相同模式出现在不同位置



特点 3

下采样不影响结果



关于深度学习的由来

- 1958：感知机 Perceptron（linear model）
- 1969：Perceptron has limitation
- 1980s：多层感知机 Multi-layer Perceptron
  - Do not have significant difference from DNN today
- 1986：反向传播算法 Backpropagation
- 1989：1 hidden layer is “good enough”, why deep?
- 2006：RBM initialization（breakthrough）
- 2011：Start to be popular in speech recognition
  - 发现GPU效果非常好
- 2012：Win ILSVRC image competition
  - 深度学习+GPU=碾压之势

Geoffrey Hinton

图灵\诺贝尔物理学奖得主

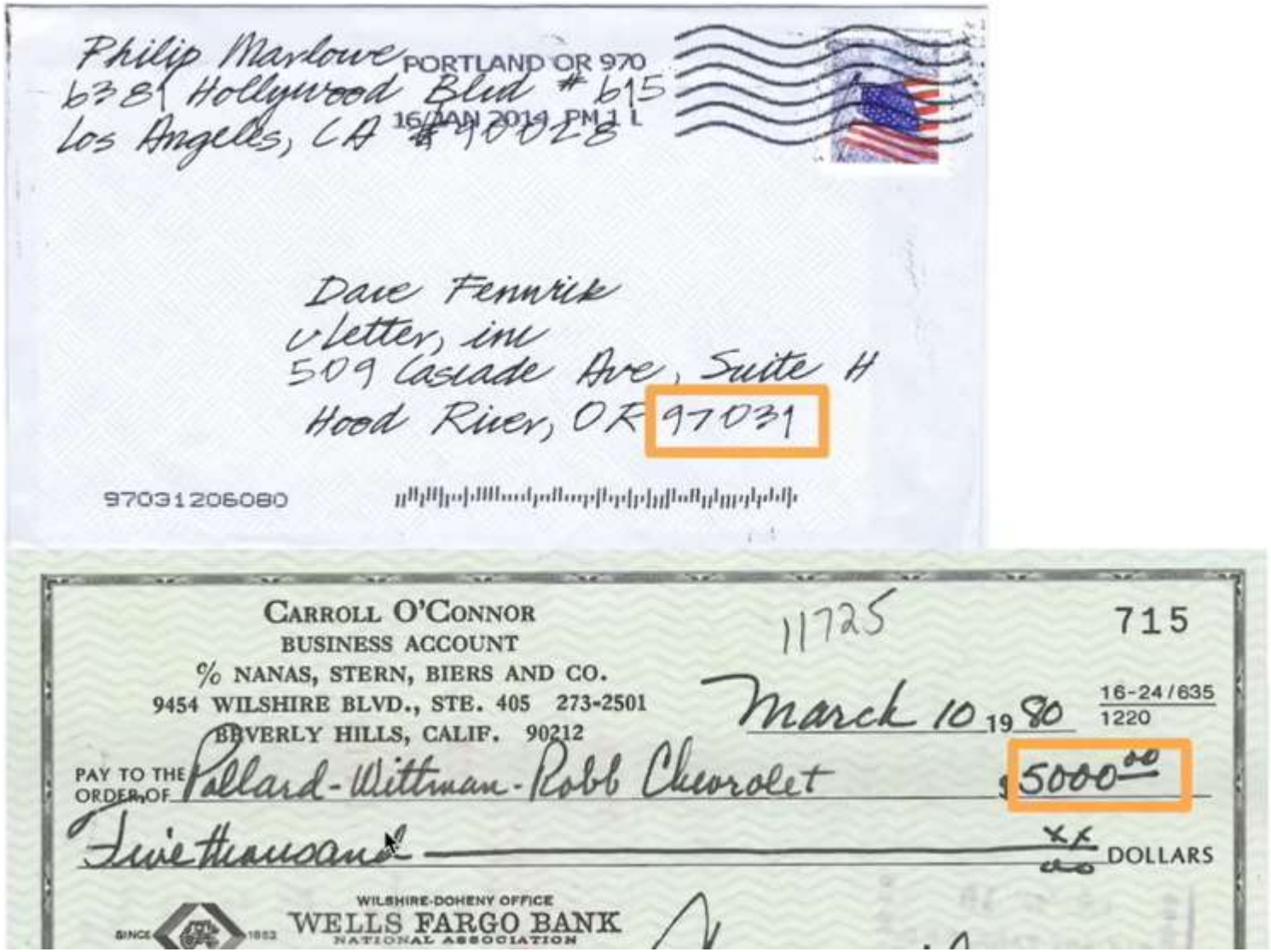
李飞飞

ILSVRC竞赛创始人

黄仁勋

Nvidia CEO

LeNet

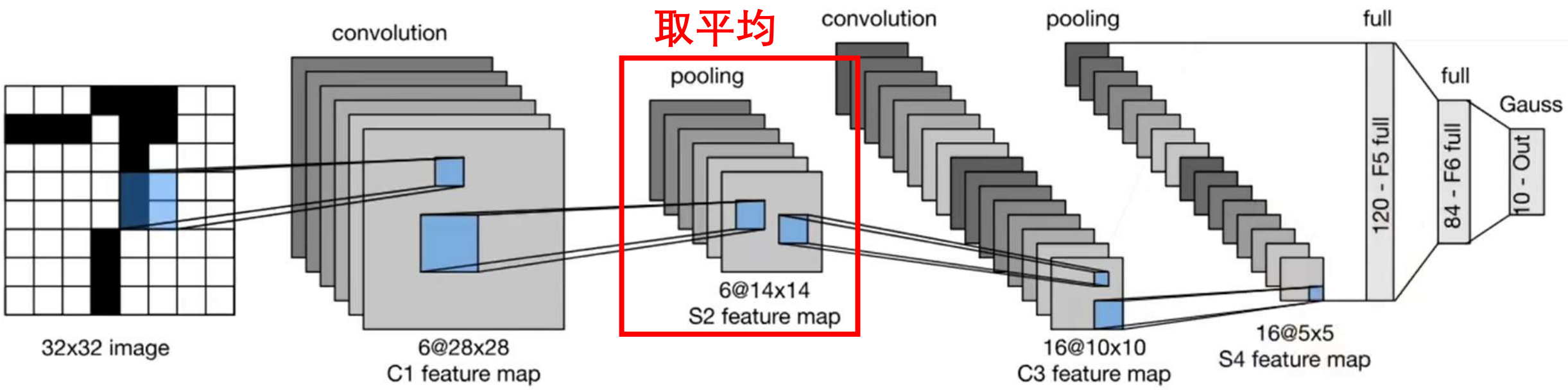


Yann LeCun

Pioneer of CNN

Yann LeCun, Corinna Cortes, and Christopher J.C. Burges. The MNIST database of handwritten digits. https://yann.lecun.com/exdb/mnist/

LeNet



- LeNet是早期成功的神经网络
- 先使用卷积层来学习图片空间信息
- 然后使用全连接层来转换到类别空间

IMAGENET （2010）

李飞飞

数据集	IMAGENET	MNIST
图片类型	自然物体的彩色图片	手写数字的黑白图片
图片大小	469 X 387	28 X 28
样本数量	1200000	60000
类数	1000	10

AlexNet (2012)

ImageNet Classification with Deep Convolutional Neural Networks

- AlexNet 赢了 2012 年 ImageNet 竞赛
- 更深更大的 LeNet
- 主要改进：
  - 丢弃法 Dropout
  - ReLu
  - MaxPooling
- 计算机视觉方法论的改变

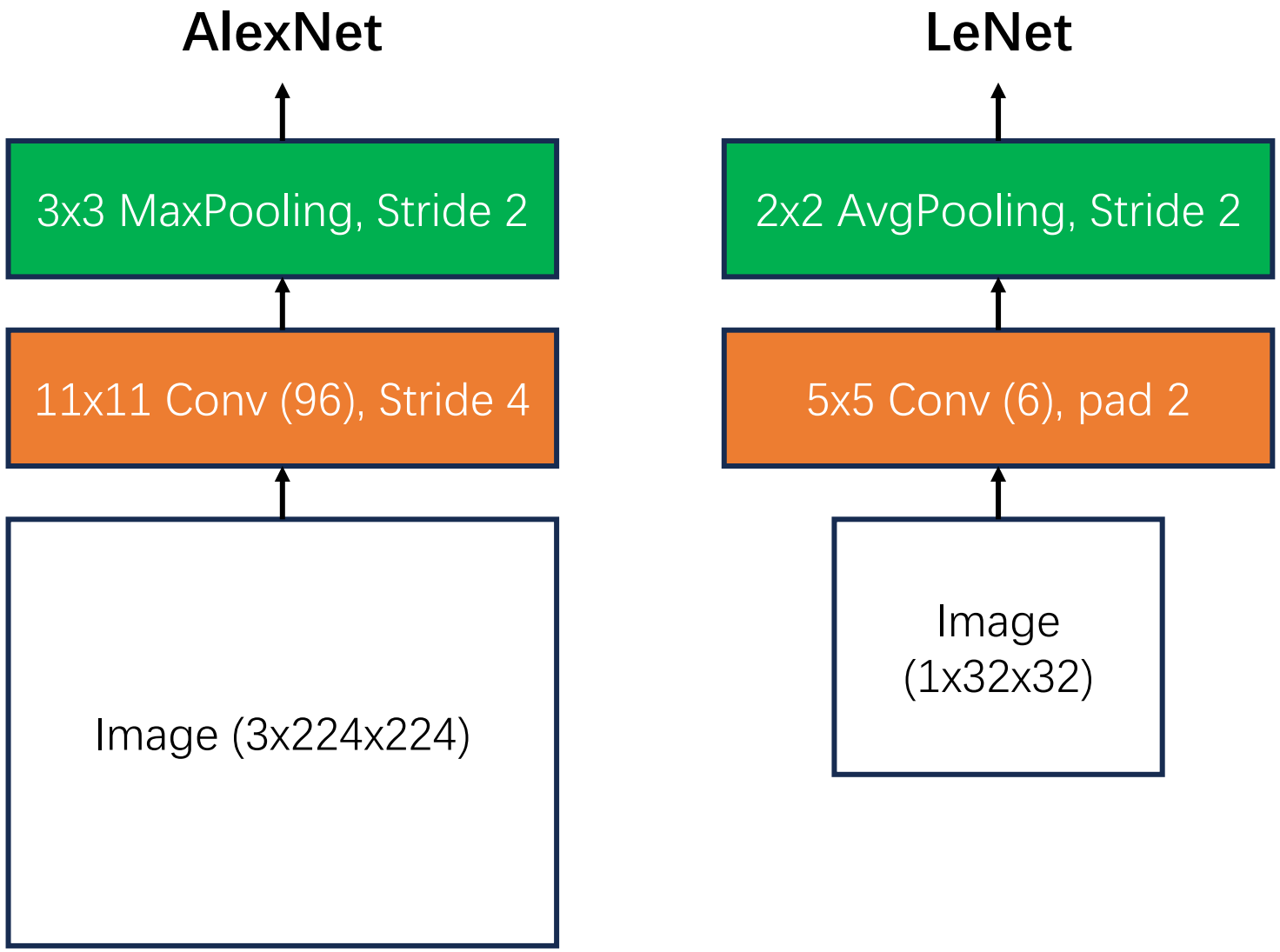
人工提取特征 → 自动提取特征

AlexNet

"Conv 1"

使用最大池化

更大的窗口和步长

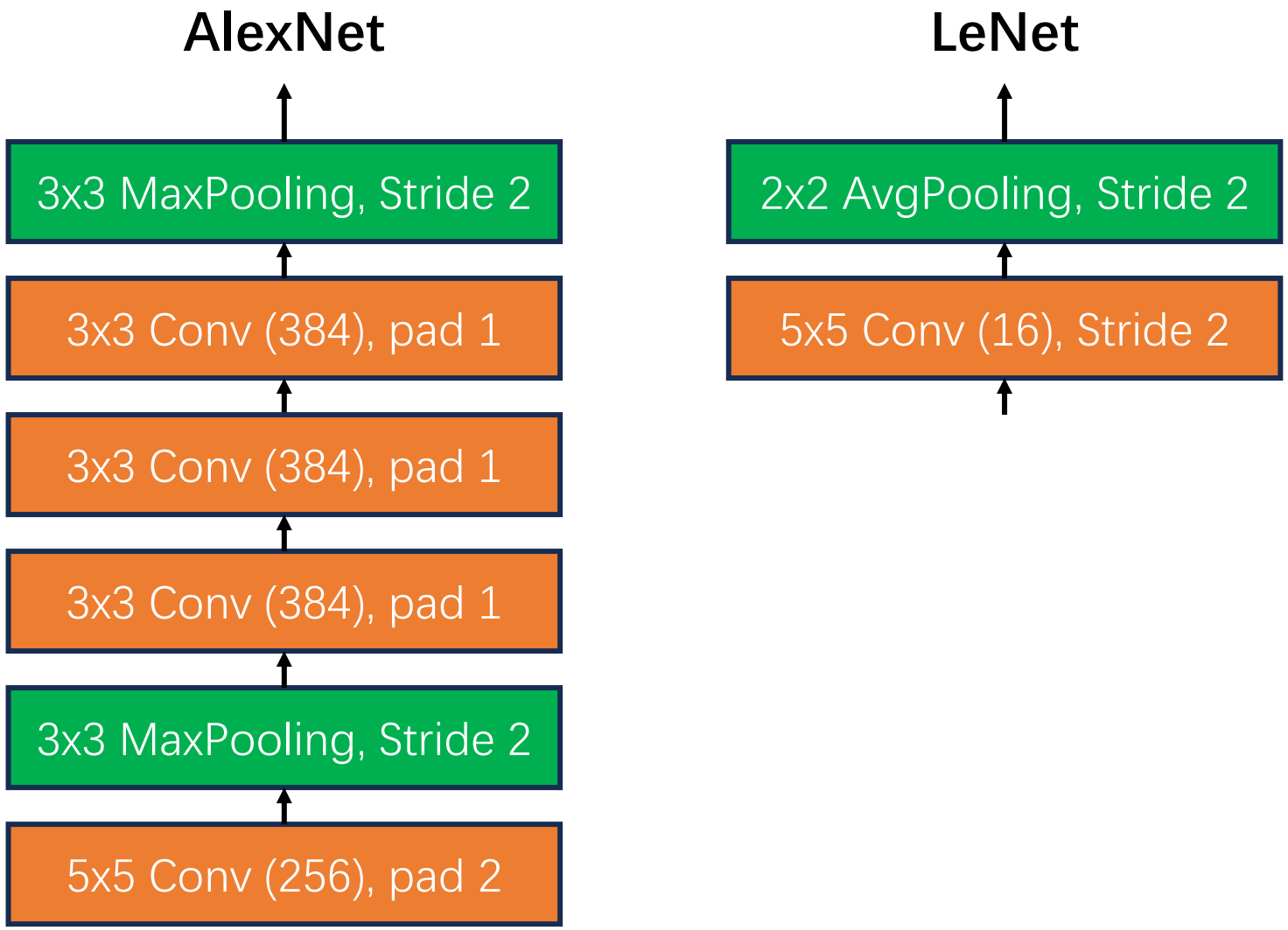


AlexNet

"Conv 2+"

连叠3层卷积层

更多的通道

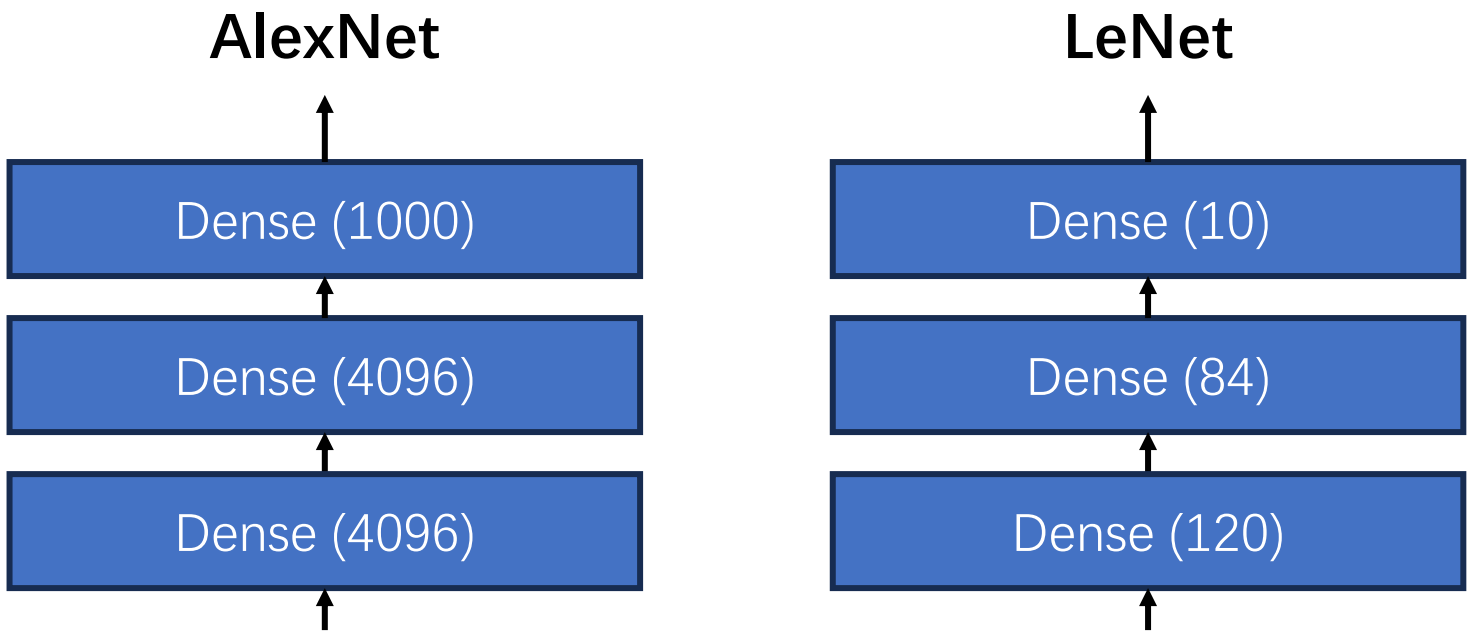


AlexNet

Fully Connecting

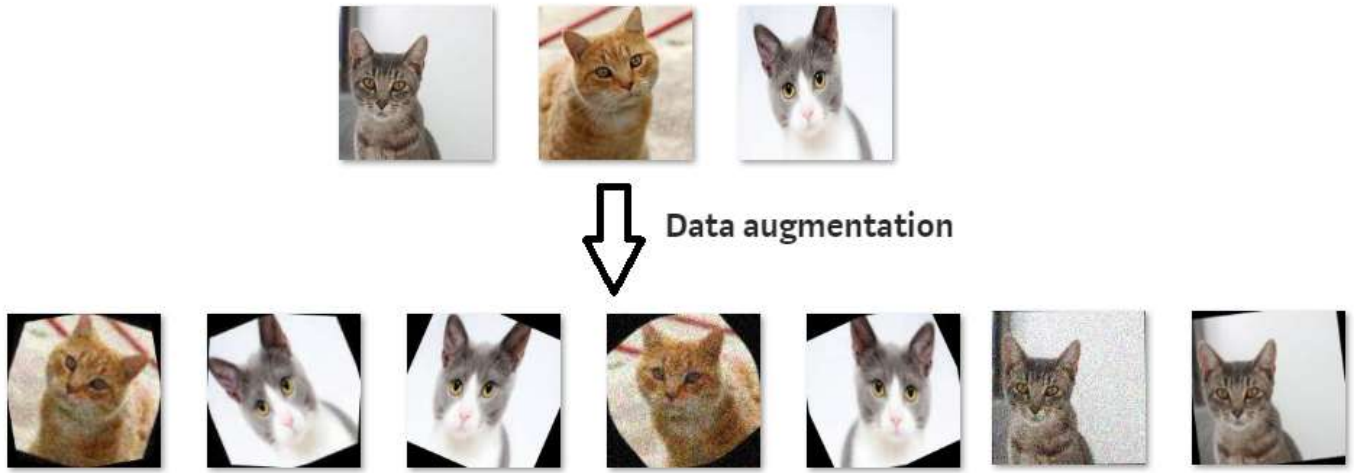
1000类输出

更多的节点数

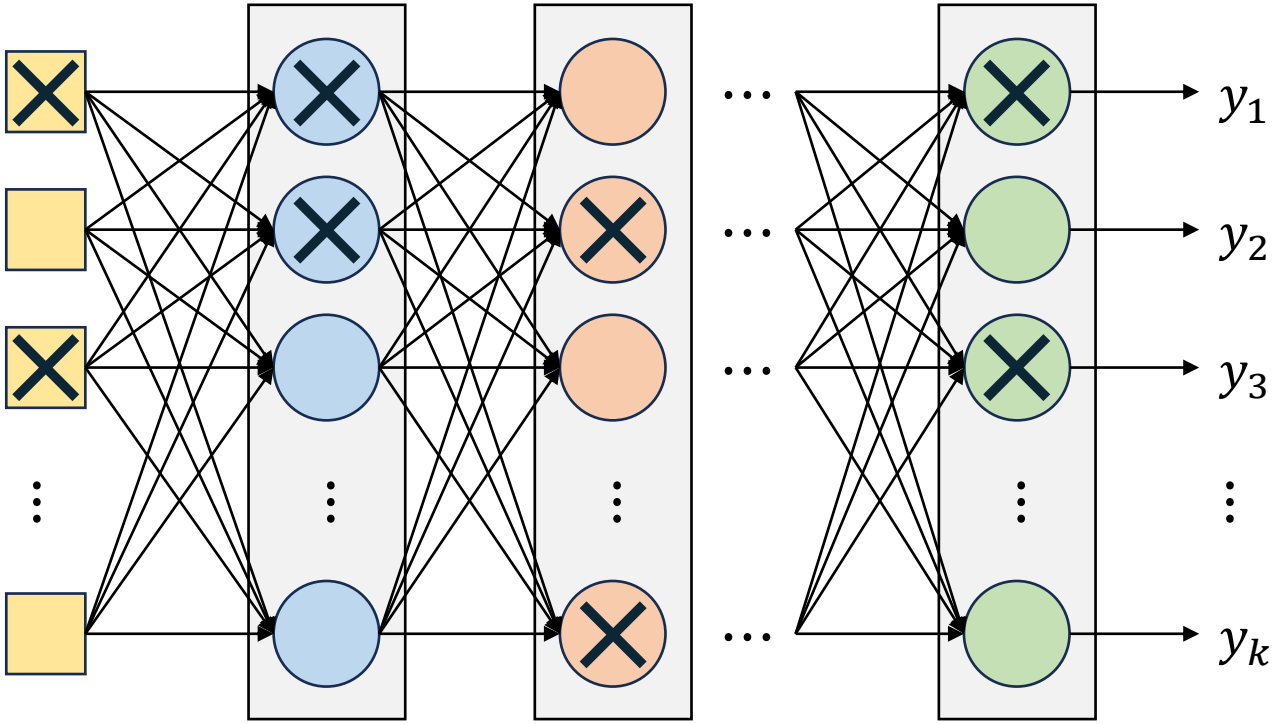


AlexNet更多的细节

- 数据增强
- 隐藏全连接层后加入了Dropout
- 激活函数从Sigmoid编导了ReLU



AlexNet更多的细节——Dropout

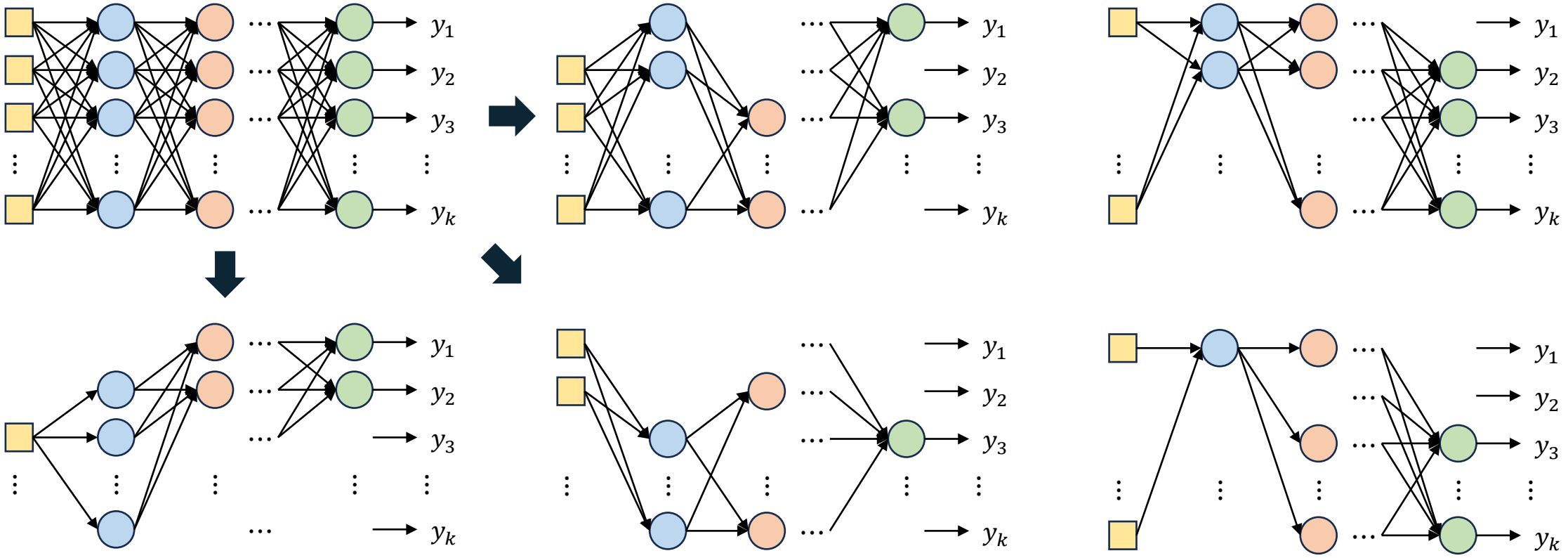


Dropout以 $p = 0.5$ 的概率丢弃特征或节点

！！！每次更新都需要重新Dropout！！！

AlexNet更多的细节——Dropout

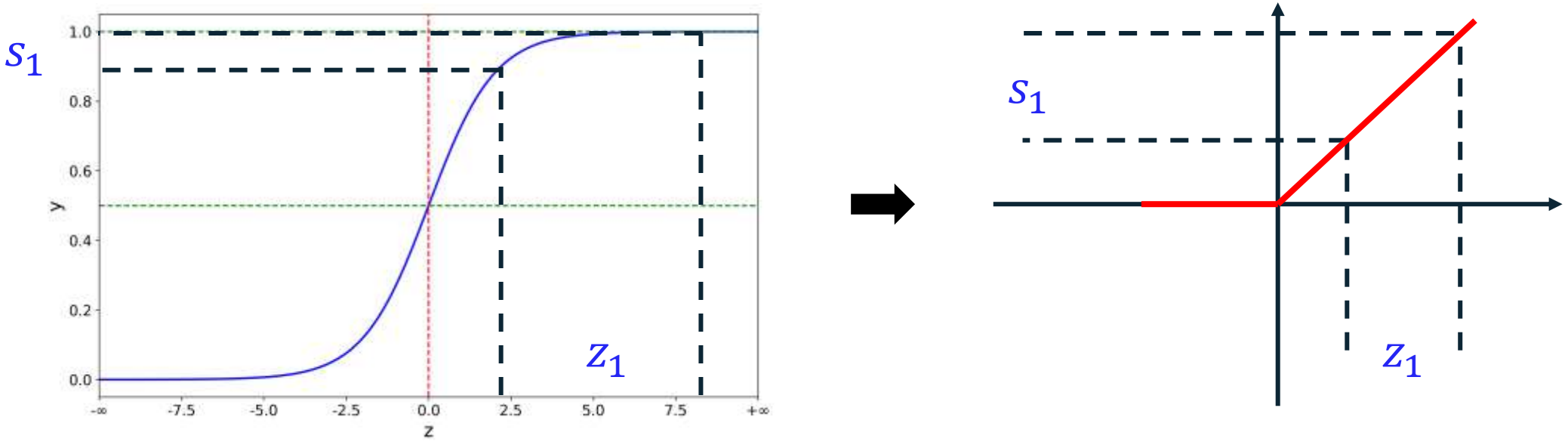
Dropout相当于训练了极多个不同的简单神经网络



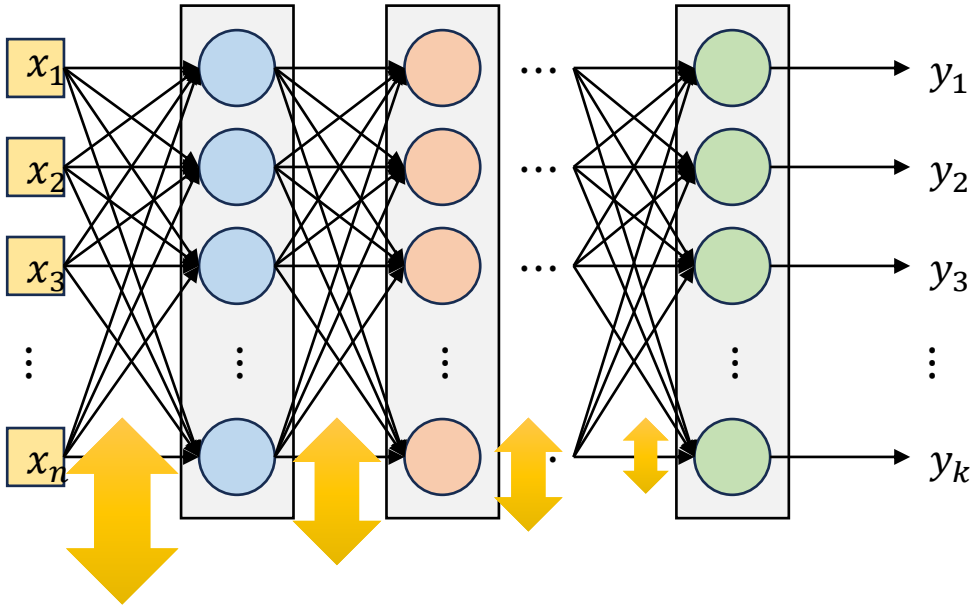
融合多个弱分类器的结果，能够达到极好的效果，并具有更强的泛化能力

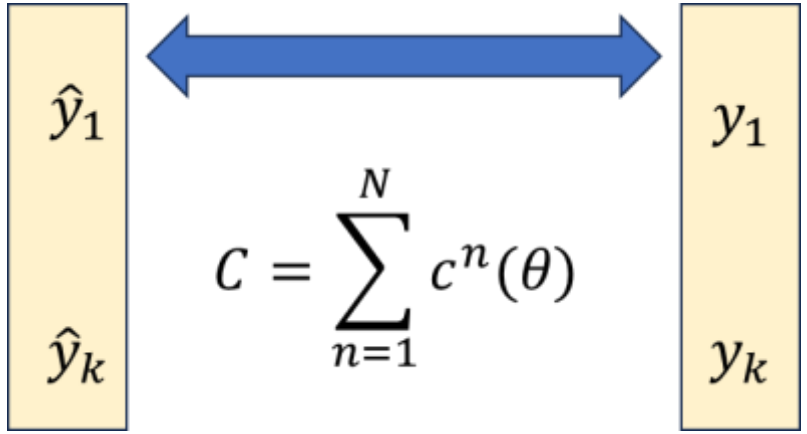
AlexNet更多的细节

- 数据增强
- 隐藏全连接层后加入了Dropout
- 激活函数从Sigmoid编导了ReLU（缓解**梯度消失**问题）



梯度消失问题

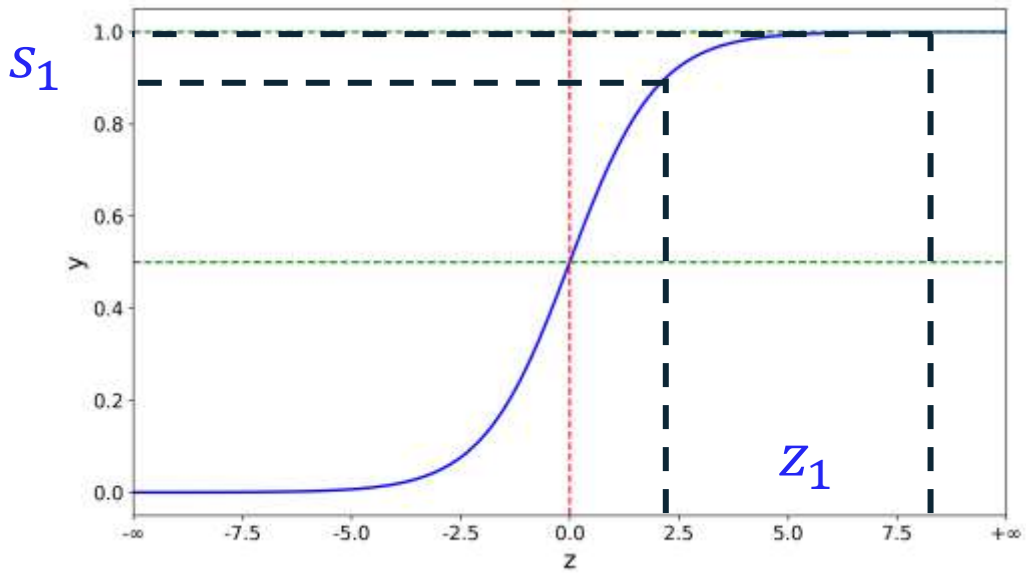




$$C = \sum_{n=1}^{N} c^n(\theta)$$

随着层数增高，低层参数对C的影响逐渐降低

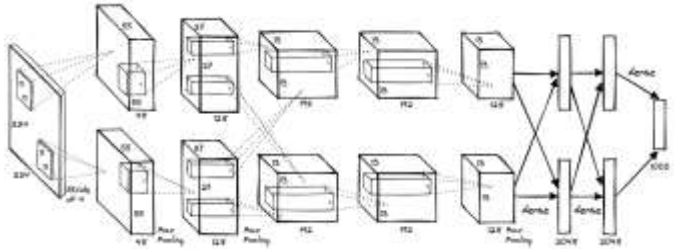
$$\frac{\partial C}{\partial w} \rightarrow \frac{\Delta C}{\Delta w}$$



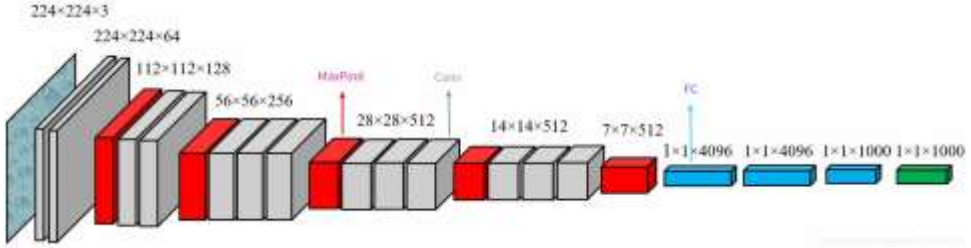
梯度消失使得深度神经网络难以训练

神经网络的能力和隐藏层数正相关

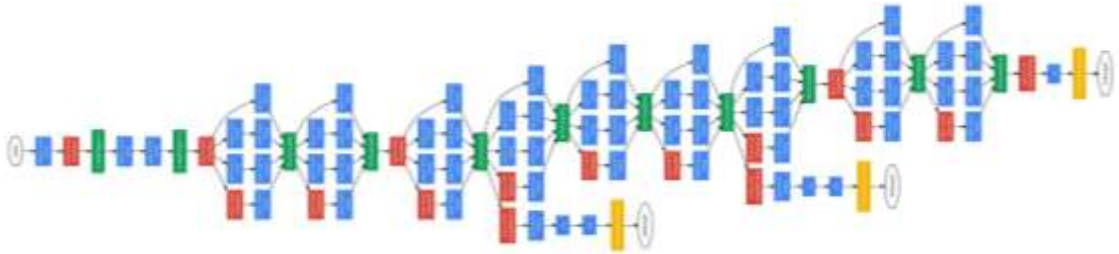
AlexNet'12
8 Layers
16.4%



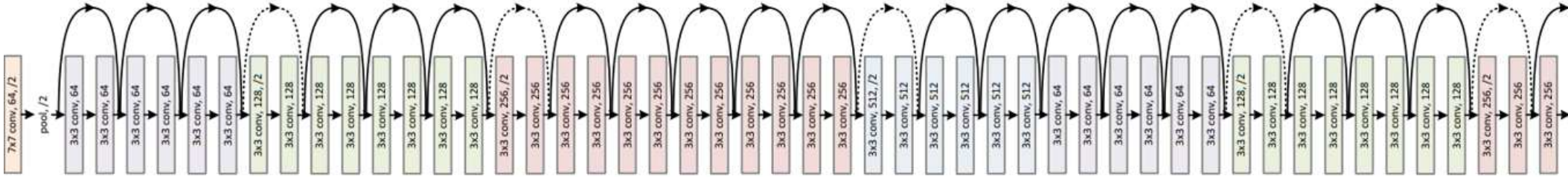
VGG'14
19 Layers
7.3%



GoogleNet'14
22 Layers
6.7%

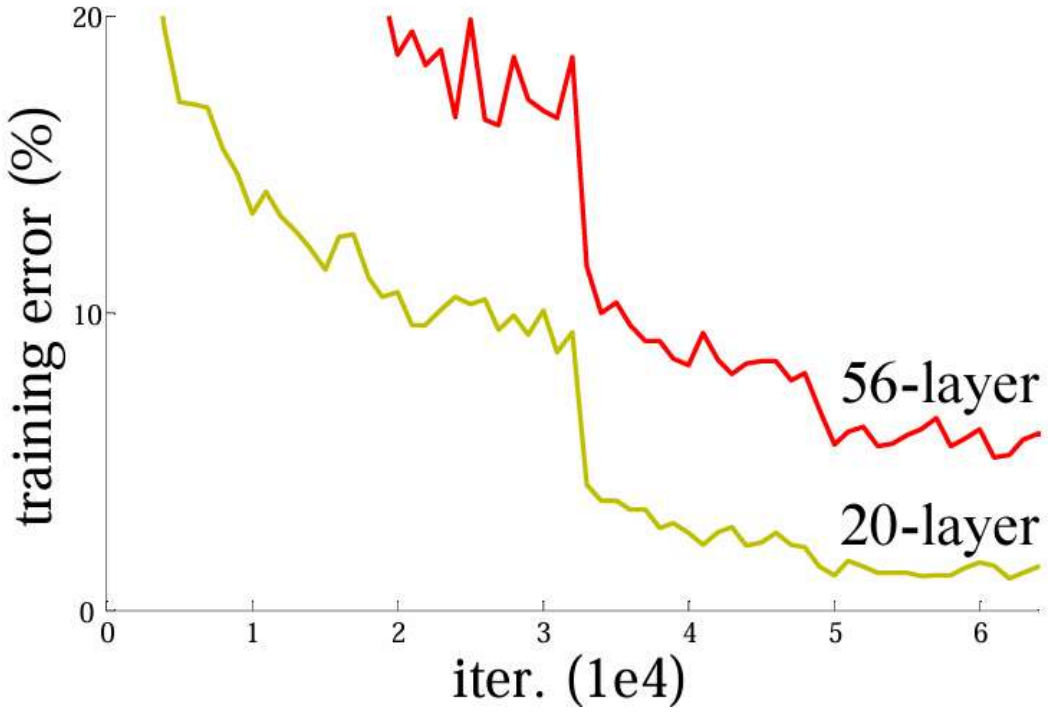


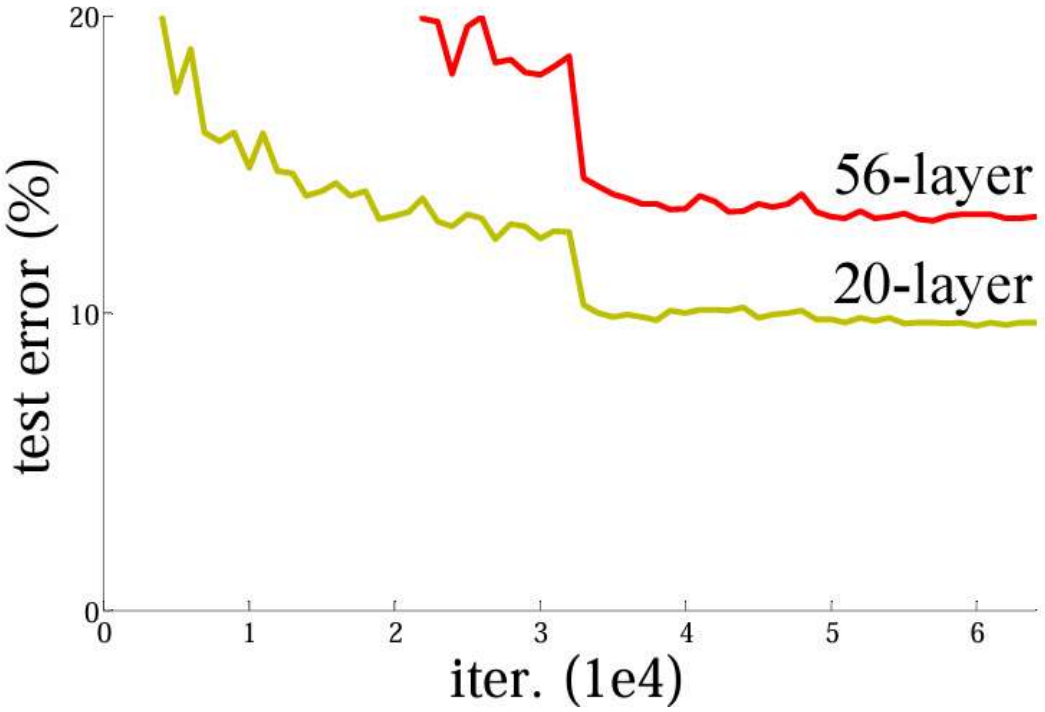
Residual Net'15
152 Layers
3.57%



ResNet让训练
真·深度神经网络
成为可能

ResNet

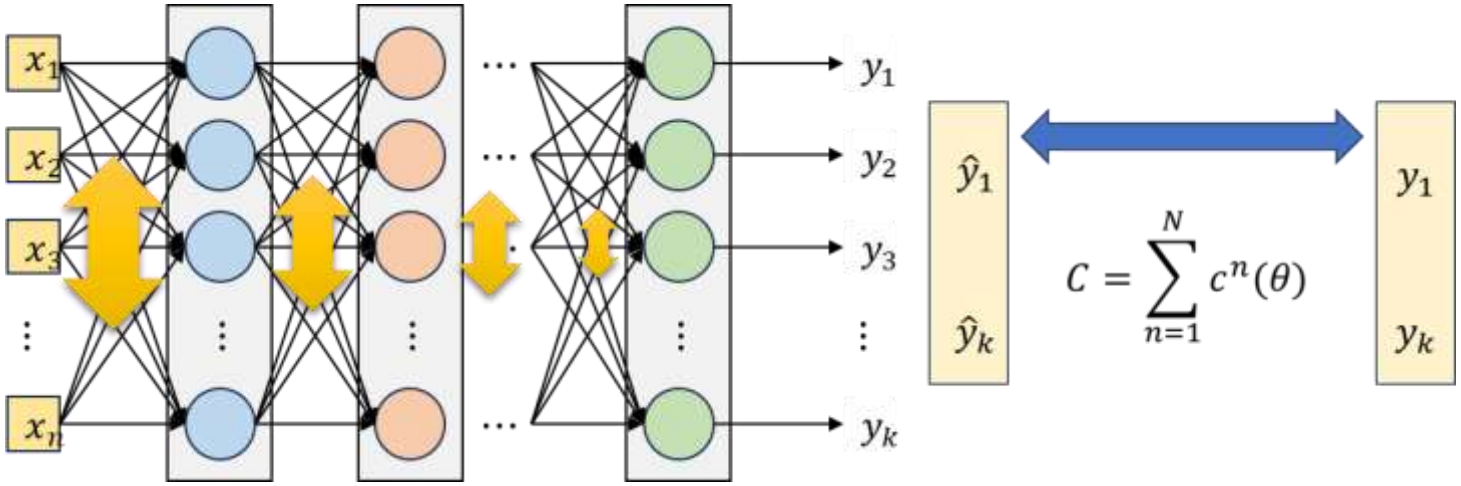




并非发生过拟合，而是训练除了偏差

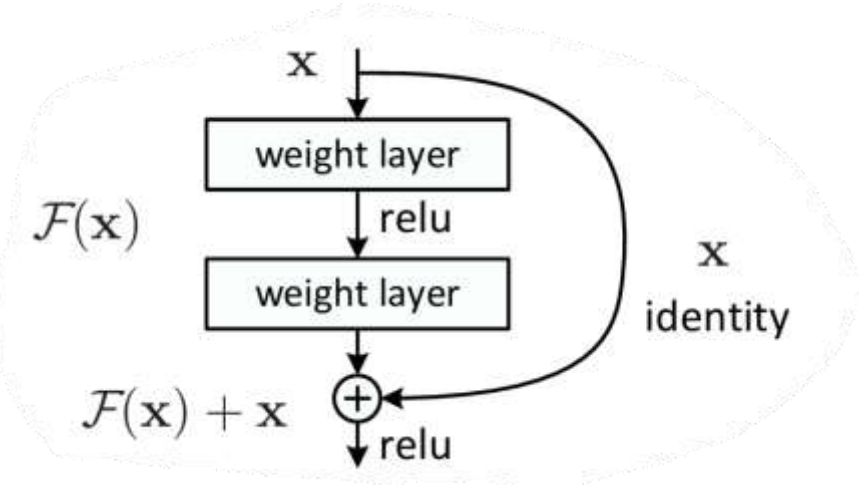
https://arxiv.org/pdf/1512.03385

ResNet

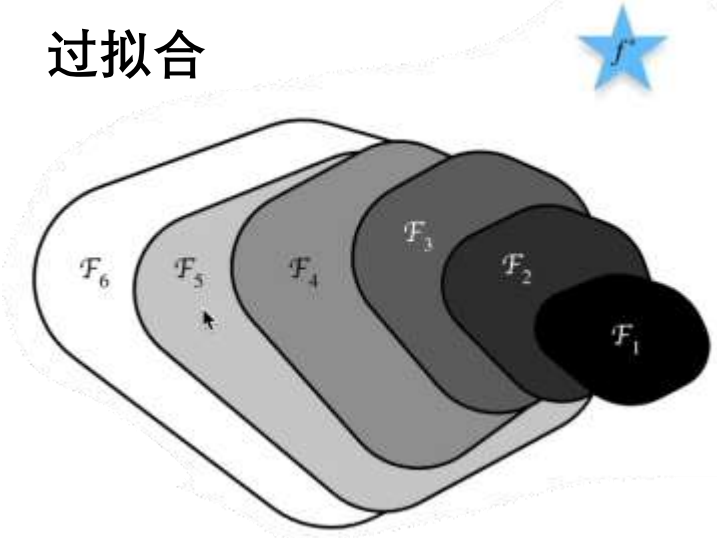


一个朴素的愿望是

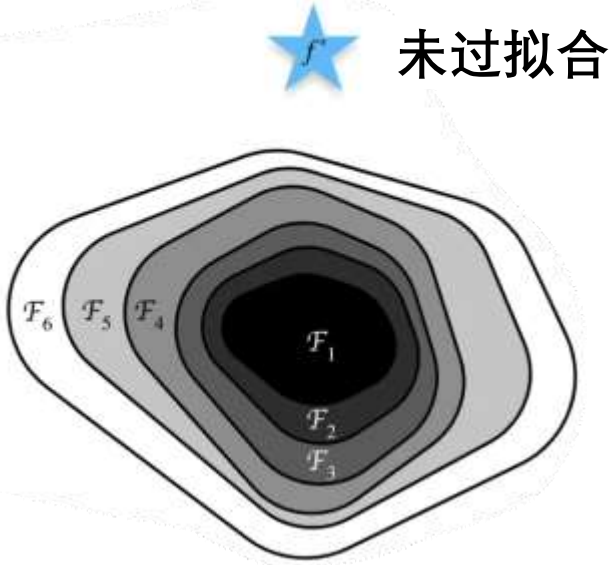
深模型至少不会比浅模型差



过拟合



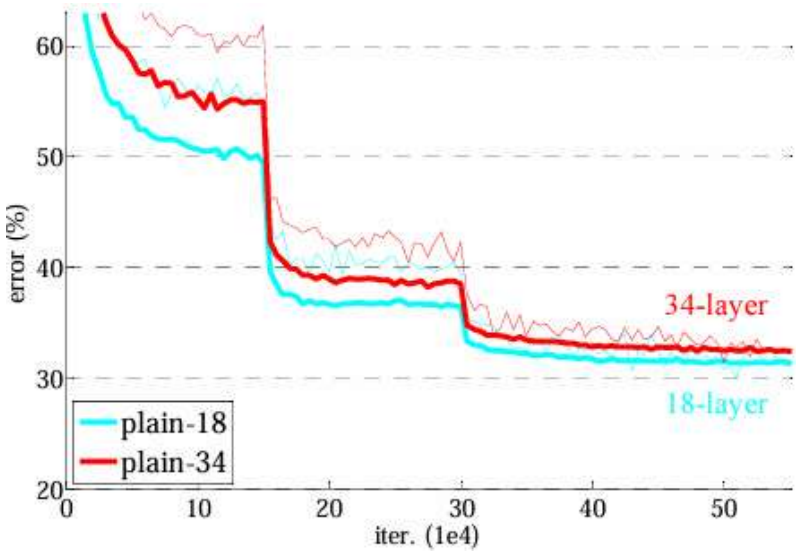
未过拟合

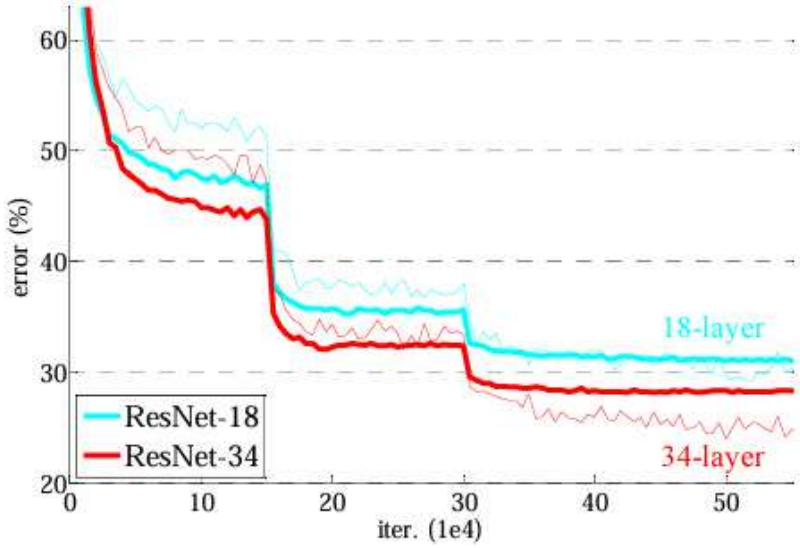


即使 F(x) 学“偏”了

x也不影响后续层的学习

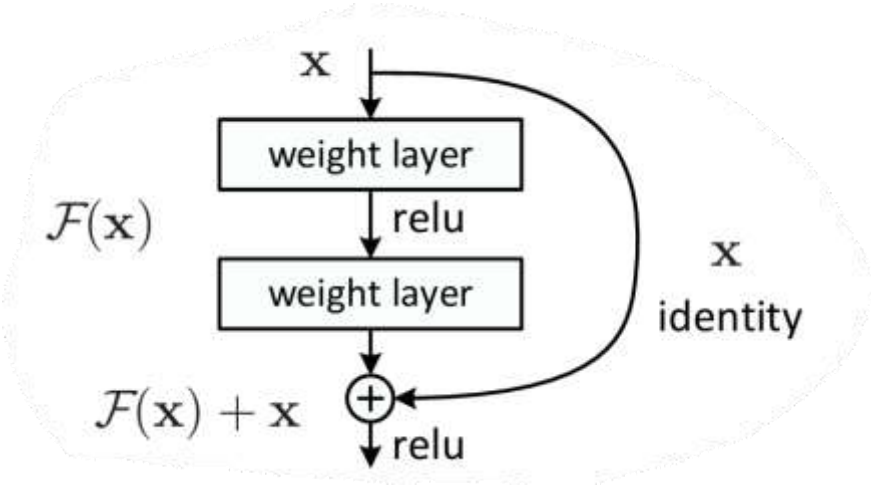
ResNet





ResNet朴素的愿望是

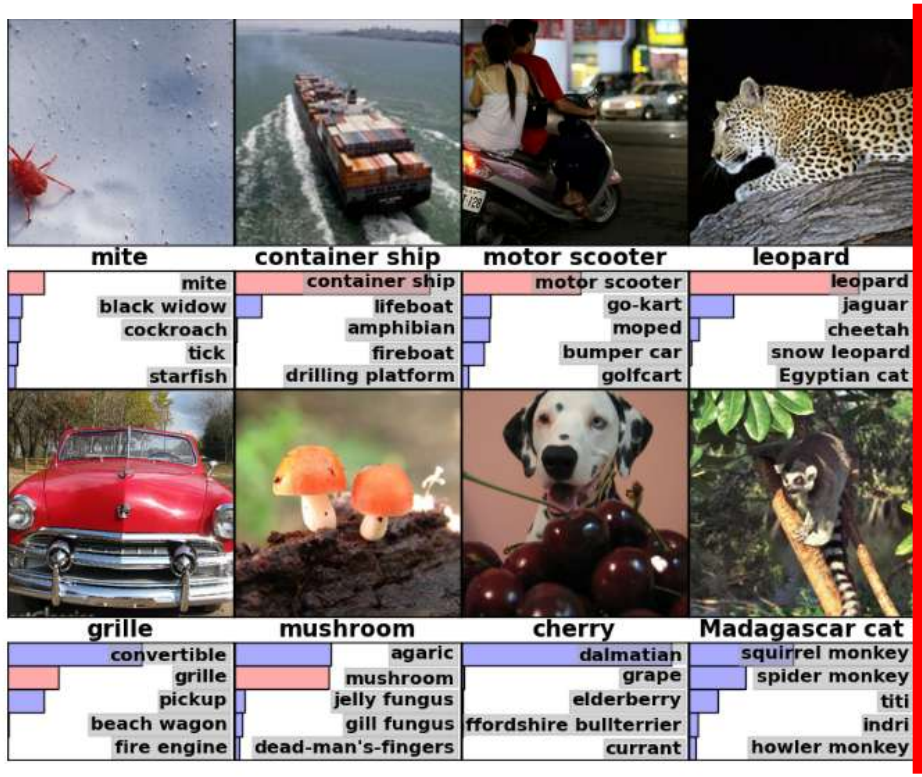
深模型至少不会比浅模型差



即使 F(x) 学“偏”了

x也不影响后续层的学习

One More Thing of AlexNet



当时没感觉但其实是最重要的发现

最后得到的特征向量能够清晰的将不同的类别区分开

小结

LeNet

AlexNet

ResNet